

Lab 8: ArrayList & Exception Handling

Problem 1: E-Commerce Product System with Abstract Classes (10 Points)

Description:

Design an online store product hierarchy using abstract classes and interfaces to handle different product types with inventory management.

Requirements:

1. **UML Class Diagram** (2 Point)
 - Abstract class: Product
 - Attributes: productId (String), price (double), stock (int)
 - Methods: updateStock(quantity) (concrete), displayDetails() (abstract)
 - Interface: Discountable
 - Method: applyDiscount(double percentage)
 - Subclasses:
 - Electronics (implements Discountable)
 - Clothing
2. **Abstract Class & Interface Implementation** (2 Point)
 - Product:
 - Constructor to set productId, price, and stock.
 - Concrete updateStock() to modify inventory.
 - Discountable:
 - Declare applyDiscount().
3. **Subclass Definitions** (2 Point)
 - Electronics:
 - Additional Attribute: warrantyMonths (int).
 - Override displayDetails() to include warranty.
 - Implement applyDiscount() (max 20% discount).
 - Clothing:
 - Additional Attribute: size (String).
 - Override displayDetails() to include size.
4. **ArrayList & Exception Handling** (2 Point)
 - Create an ArrayList<Product> to store 3 products (2 Electronics, 1 Clothing).
 - Use try-catch to handle IllegalArgumentException in applyDiscount() if discount > 50%.
5. **Testing** (2 Point)
 - Add products:
 - Electronics: ("E1", 999.99, 10, warranty=12).
 - Clothing: ("C1", 49.99, 5, size="M").
 - Apply 15% discount to Electronics, print details.
 - Attempt 60% discount (catch exception).

Expected Output:

Electronics ID: E1 | Price: 849.99 | Warranty: 12 months
Clothing ID: C1 | Size: M | Price: 49.99
Error: Discount cannot exceed 50%!